

Theory Assignment I

Automata Theory Monsoon 2026, IIT Hyderabad

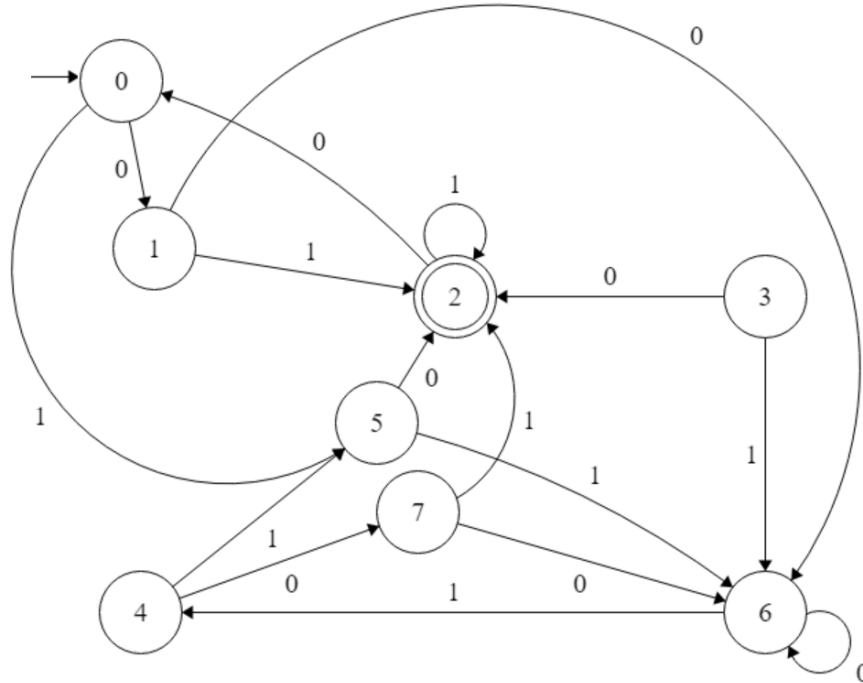
August 7, 2026

Total Marks: 40 points

Due date: **23rd August, 2026**

General Instructions: All symbols have the usual meanings (example: $\mathbb{R}$ is the set of reals, $\mathbb{N}$ the set of natural numbers, and so on). DFA stands for deterministic finite automata. NFA stands for non-deterministic finite automata. a^* is the Kleene Star operation.

- [2 + 2 points] (a) Let $\Sigma = \{a, b\}$. For each $k \geq 1$, let C_k be the language consisting of all strings that contain an a exactly k places from the right-hand end. Describe an NFA with $k + 1$ states that recognizes C_k , both in terms of a state diagram and a formal description.
(b) Consider the languages C_k defined in the previous problem. Prove that for each k , no DFA can recognize C_k with fewer than 2^k states. [CO1, CO2, CO3, CO4]
- [4 points] The Hamming distance between two words of equal length is the number of positions at which they differ. Prove that for every regular language L and every constant k , the set of words at a distance at most k from a word in L is regular. [CO1, CO2, CO3]
- [3 points] Minimize the DFA given below: [CO1, CO3]



4. [3 points] Let $\Sigma = \{a\}$ be a single-letter alphabet. Prove that a language $L \subseteq \Sigma^*$ is regular if and only if the set of the lengths of its words, $S = \{|w| \mid w \in L\}$, is ultimately periodic.

Definition: A set of natural numbers $S \subseteq \mathbb{N}$ is ultimately periodic if there exists a threshold integer $N \geq 0$ and a period integer $p \geq 1$ such that for all integers $n \geq N$:

$$n \in S \iff n + p \in S$$

In other words, past a certain point N , the inclusion of numbers in the set repeats in a perfect cycle of length p .

[CO1, CO3]

5. [3 points] Consider the language $L = \{a^i b^j \mid i \neq j\}$.
- Prove that L is not regular with the help of the pumping lemma
 - Provide a much shorter proof that L is not regular using only closure properties and the known non-regular language $L_{eq} = \{a^n b^n \mid n \geq 0\}$.

[CO1, CO2, CO3, CO4]

6. [3 points] Let

$$\Sigma_3 = \left\{ \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \dots, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}.$$

Σ_3 contains all size-3 columns of 0s and 1s. A string of symbols in Σ_3 gives three rows of 0s and 1s. Consider each row to be a binary number written with the *least significant bit first* (i.e. the leftmost column is the units bit) and let

$$B = \{w \in \Sigma_3^* \mid \text{the bottom row of } w \text{ is the sum of the top two rows}\}.$$

For example,

$$\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \in B, \quad \text{but} \quad \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \notin B.$$

Show that B is regular. [CO1, CO2, CO3]

7. [8 points] Prove that the following languages are not regular:
- $\{a^{2^n} : n \in \mathbb{N}\}$
 - $\{a^p : p \text{ is a prime number}\}$
 - $\{a^i b^j : \gcd(i, j) = 1\}$
 - $\{\text{bin}(p) : p \text{ is a prime number}\}$

[CO1, CO3, CO4]

8. [1+1+1 points] Prove that the following grammars are ambiguous by providing a string with two different leftmost/rightmost derivations, or two different parse trees

(a)

$$\begin{aligned} S &\rightarrow aAB, \\ A &\rightarrow bBb, \\ B &\rightarrow A \mid \lambda. \end{aligned}$$

(b)

$$S \rightarrow SS \mid \lambda \mid aSb \mid bSa.$$

(c)

$$S \rightarrow aSb \mid SS \mid \lambda.$$

[CO1, CO2, CO3]

9. [2 + 2 + 1 points] Construct a PDA for each of following languages.

(a)

$$L = \{a^m b^n \mid m \leq n \leq 2m\}.$$

(b)

$$L = \{a^i b^j c^k \mid i = j \text{ or } j = k\}.$$

- (c) If the constructed a PDA which accepts by final state, convert it into a PDA (for any of the two above questions) which accepts by empty stack and vice-versa

[CO1, CO2]

10. [4 points] If A is any language, let $A_{1/3-1/3}$ be the set of all strings in A with their middle thirds removed so that $A_{1/3-1/3} = \{xz \mid \text{for some } y, |x| = |y| = |z| \text{ and } xyz \in A\}$. Show that, if A is regular, then $A_{1/3-1/3}$ is not necessarily regular.

[CO1, CO3, CO4]

END OF ASSIGNMENT
